

Mosap Abdel-Ghany

AI Engineer | LLM Systems, RAG & Agentic AI

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Professional Summary

AI Engineer specializing in production LLM systems, Retrieval-Augmented Generation (RAG), and agentic AI. Built and deployed a multi-tenant enterprise AI analytics platform combining Text-to-SQL, hybrid semantic retrieval (BM25 + vector embeddings), and agentic reasoning over a 3.9GB production database. Experienced across the full AI system lifecycle: LLM orchestration, prompt engineering, backend architecture with FastAPI and async Python, automated testing, and Docker-based deployment.

Education

B.Sc. in Artificial Intelligence | Helwan International Technology University (HITU)

2022 – 2026

Technological College, Cairo | Graduation: 2026 | Grade: Excellent

Work Experience

AI Engineer | Core Business Solutions (CBS)

November 2025 – September 2026

- Architected Eliara, a multi-tenant enterprise AI analytics platform enabling natural-language Text-to-SQL queries over a 3.9GB production database (378 tables, 464 views), serving multiple client companies with isolated data, caches, and audit trails.
- Designed a two-model LLM architecture separating orchestration from constrained SQL generation, layered with an AST-based SQL validator and read-only execution safeguards, significantly reducing unsafe and malformed SQL execution.
- Built a hybrid semantic retrieval engine (BM25 + vector embeddings) for schema discovery, achieving 100% top-3 retrieval accuracy on a 90-question benchmark.
- Implemented agentic reasoning workflows for open-ended investigative queries, real-time streaming responses, and a rule-based business alert system operating independently of the LLM.
- Deployed the platform to production with Docker, automated backups, health monitoring, structured audit logging, and a 390+ test automated suite covering security, isolation, and regression scenarios.
- Diagnosed and resolved 8 root-cause bugs in a separate FastAPI + SQLite vehicle valuation pricing API, correcting data-cleaning, comparable-selection, and price-adjustment logic to fix inaccurate price outputs.

Python Instructor | TM Academy

November 2024 – November

2025

- Delivered a structured Python curriculum from fundamentals through OOP and file handling, with hands-on projects and assessments.
- Provided individualized mentoring and feedback that improved learners' practical coding and problem-solving skills.

Internships & Trainings

Generative AI Trainee | Digital Egypt Pioneers Initiative (DEPI)

November 2025 – July 2026

- Gaining advanced training in Generative AI concepts including large language models, prompt engineering, RAG pipelines, and fine-tuning techniques.
- Applying generative models to practical use cases including text generation, summarization, and intelligent document processing.
- Working on hands-on projects integrating LLMs with vector databases and agentic frameworks to build production-ready GenAI solutions.

AI for Business – 120-Hour Program | National Telecommunication Institute (NTI)

Oct – Dec 2025

- Applied neural networks for customer segmentation, demand forecasting, and churn prediction aligned to KPIs.
- Evaluated models using revenue-centric metrics such as cost savings, churn impact, and ROI, not accuracy alone.
- Translated model outputs into executive-ready insights for stakeholder presentations.

Machine Learning – 120-Hour Program | National Telecommunication Institute (NTI)

Aug – Sep 2025

- Mastered supervised and unsupervised learning algorithms including ensemble methods (Random Forest, LightGBM, XGBoost).
- Applied advanced techniques in model selection, validation, and performance tuning on real datasets.

Data Science Trainee | Digital Egypt Pioneers Initiative (DEPI)

October 2024 – May 2025

- Conducted EDA, data preprocessing, and statistical analysis using Python (Pandas, NumPy) and SQL on large-scale datasets.
- Built and evaluated predictive models (Logistic Regression, Random Forest) with real-world business applications.
- Developed interactive Streamlit and Plotly dashboards to communicate insights to non-technical stakeholders.

Projects

Pathora — AI-Powered Career Assessment Platform — Graduation Project, Team Lead

- Led a nine-persons team building a full-stack platform (FastAPI, async SQLAlchemy 2.0, SQLite backend; React 19, Vite frontend) for CV/JD-based job-title prediction, personalized exam generation, and course recommendations.
- Designed a dual-provider LLM layer (Ollama locally or Hugging Face Inference API) switchable via a single configuration flag, with model choices tuned for constrained hardware (4GB VRAM).
- Built an LLM call tracing system to debug and resolve JSON malformation, scoring-logic, and integration issues across the AI pipeline.

Multilingual Sentiment Analysis — Graduation Project, Digital Egypt Pioneers Initiative, May 2025

- Built an NLP pipeline for TripAdvisor and Quora Egypt datasets handling Arabic and English text with preprocessing and data augmentation (back-translation).
- Trained and benchmarked Logistic Regression, SVM, LSTM, and BERT models; visualized insights interactively with Plotly and Streamlit.

Sola POS — Freelance, Desktop Point-of-Sale System

- Built a commercial, production-ready desktop POS for restaurants and cafes: Electron, React, and TypeScript frontend with a FastAPI and PostgreSQL backend in a Clean Architecture monorepo.
- Delivered the full order lifecycle (hold, resume, cancel), multi-method payments, cash-shift management, role-based access, and a reporting suite (sales, staff, cash summaries).
- Engineered a fully self-contained cashier-machine deployment by bundling PostgreSQL and a PyInstaller-packaged backend directly into the Electron app, removing any Docker or setup dependency for end users.

Customer Churn Prediction Dashboard

- Deployed a Streamlit-based Random Forest model achieving 91% accuracy to predict churn risk and support proactive retention decisions.
- Outperformed manual methods by 28%, enabling businesses to reduce acquisition costs by 5–25x through targeted intervention.

Technical Skills

AI & LLM Engineering: LLM Applications, Retrieval-Augmented Generation (RAG), Agentic AI, LLM Orchestration, Prompt Engineering, Text-to-SQL, Semantic Search, Vector Search, Hybrid Search (BM25 + Vector), Embeddings

Programming: Python, SQL, TypeScript

Machine Learning & NLP: Scikit-learn, PyTorch, TensorFlow, NLP, Supervised & Unsupervised Learning, Deep Learning

Backend & AI Systems: FastAPI, SQLAlchemy, REST APIs, Async Python, Streaming Systems (SSE), Multi-Tenant Architecture, pytest

Databases: PostgreSQL, SQL Server, SQLite, Firebase/Firestore

Infrastructure & Tools: Docker, Git/GitHub, Jupyter Notebook, Ollama, Hugging Face Inference API

Volunteering Activities

AI & Machine Learning Instructor | RTC

July 2024 – Present

- Mentored students in applied ML techniques using Python, Scikit-learn, and Pandas; fostered collaborative learning through Q&A and structured feedback.

OC Team Leader | RTC

March 2024 – Present

- Oversaw team progress tracking and coordinated timely interventions to keep learning milestones on schedule; developed leadership and mentoring competencies.

HR Team Member | Sunrise Academy

August 2024 – May 2025

- Managed recruitment cycles, task delegation, and talent follow-up; strengthened skills in talent management and organizational coordination.

Certifications

DataCamp (2025): Machine Learning Fundamentals, Data Analysis in Python Track, Python Toolbox

365 Data Science (2024): Data Scientist Career Track, Deep Learning, Machine Learning, Power BI, SQL, Mathematics & Probability/Statistics for Data Analysis

Kaggle (2023): Python

Languages

Arabic: Native / Mother Tongue

English: Intermediate (B1)

Reference: Available upon request